



Dataset Preparation & Upload Guidelines

Ensure accurate and reliable DDoS detection results by following this guide.

v2.0 Updated

■ Supports: CSV · PCAP · PCAPNG · CAP

■ Models: Random Forest · Logistic Regression

1 Supported Input Formats

The DDoS Analyzer accepts **four file formats**. CSV files contain pre-processed flow data. PCAP, PCAPNG, and CAP files are raw packet captures that the system converts automatically — no external tools needed.

Format	Extension	Source Tool	Conversion	Best For
Flow CSV	.csv	CICFlowMeter	✓ Ready to use	Research · labeled datasets
Packet Capture	.pcap	Wireshark · tcpdump · hping3	■ Auto-converted	Live traffic · attack simulations
Next-Gen Capture	.pcapng	Wireshark (default save)	■ Auto-converted	Modern Wireshark captures
Legacy Capture	.cap	tcpdump · older tools	■ Auto-converted	Legacy scripts · older capture files

2 Creating Your Own Dataset with CICFlowMeter

CICFlowMeter is the official traffic flow generation tool developed by the Canadian Institute for Cybersecurity. It converts raw network traffic into the 80-feature flow format that our ML models are trained on.

Download from the official source:

<https://www.unb.ca/cic/research/applications.html>

01

Download CICFlowMeter

Get the official tool from [unb.ca/cic/research/applications.html](https://www.unb.ca/cic/research/applications.html)

02

Capture Network Traffic

Record traffic using Wireshark, tcpdump, or hping3 — save as .pcap or .pcapng

03

Run CICFlowMeter in Offline Mode

Open CICFlowMeter → set Pcap dir to your file → set Output dir → click OK

04

Upload the Generated CSV

The output CSV has 80 standardised flow features — upload directly to the analyzer

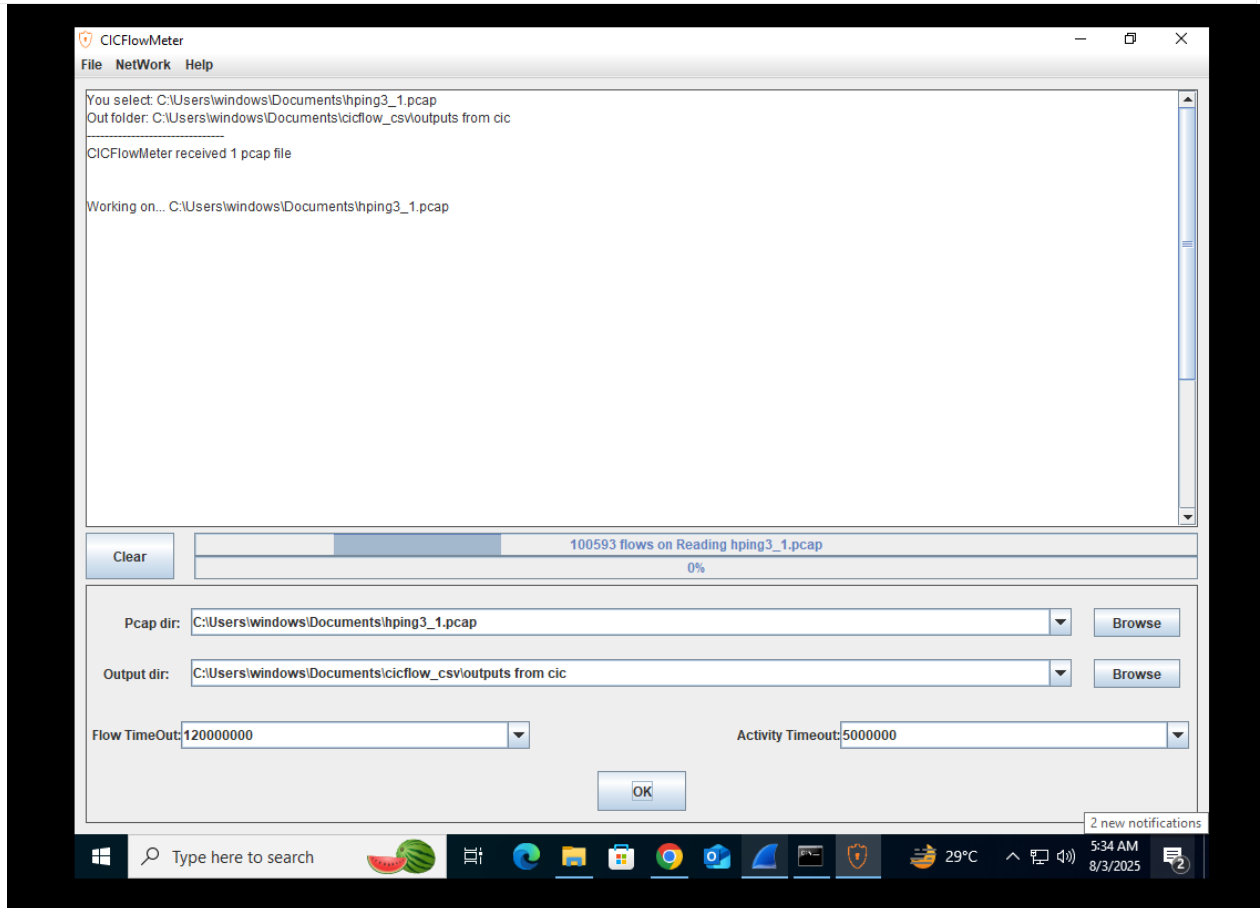


Figure: CICFlowMeter offline mode — converting a PCAP file into a CSV network traffic dataset.

3

Direct PCAP / PCAPNG / CAP Upload**NEW**

Skip CICFlowMeter entirely. Upload your raw packet capture file directly. The system uses a built-in Scapy-powered extractor to compute all 80 CICFlowMeter-compatible features automatically — and runs the full analysis in one step.

Supported raw capture formats:

Format	Extension	Typical Source	Notes
PCAP	.pcap	Wireshark, tcpdump, hping3	Most widely supported format
PCAPNG	.pcapng	Wireshark default save	Supports multiple interfaces and metadata
CAP	.cap	tcpdump, older tools	Same binary structure as PCAP

■ **Accuracy note:** The built-in extractor closely approximates CICFlowMeter but is not identical to the official Java implementation. For highest accuracy in research, convert using CICFlowMeter and upload the resulting CSV.

4 CSV Format Requirements

When uploading a **CSV file**, it must follow the CICFlowMeter 80-feature column schema. The system handles minor column name variations (spaces, capitalisation) automatically.

Requirement	Detail	If Missing / Failed
80 flow features	Standard CICFlowMeter output columns	Missing columns filled with 0 — reduced accuracy
Label column (optional)	Values: BENIGN or DDoS	Predictions still work — accuracy metrics hidden
No corrupt rows	Rows with >20% NaN values auto-dropped	Corrupt rows removed automatically
UTF-8 encoding	Standard CSV text encoding	Upload may fail

5 Labeled vs. Unlabeled Datasets

Results shown in the dashboard depend on whether your upload contains a **Label column**.

Understanding the difference helps you interpret the output correctly.

✓ Labeled Dataset

Label column present — values: BENIGN / DDoS

- Real live accuracy percentage
 - Actual confusion matrix from your data
 - Precision / Recall / F1 scores
 - Full RF vs LR model comparison
- Ideal for: Researchers and model testing*

■ Unlabeled Dataset

No Label column — real-world live traffic

- BENIGN / DDoS predicted flow counts
- Protocol breakdown: TCP vs UDP
- Top targeted ports analysis
- AI-generated security recommendations

Ideal for: SOC teams and live analysis

6 Pre-Upload Checklist

✓	File is one of: .csv .pcap .pcapng .cap
✓	File size is under 500 MB
✓	CSV only: columns match CICFlowMeter 80-feature schema
✓	CSV only: file is UTF-8 encoded
✓	PCAP/PCAPNG/CAP: captured on standard Ethernet/IP network
✓	PCAP/PCAPNG/CAP: file opens without errors in Wireshark
■	Optional: add a Label column (BENIGN / DDoS) to see accuracy metrics



Optional: use official CICFlowMeter for highest CSV precision

7 About the Detection Models

Random Forest

n_estimators=200 · max_depth=20 ·
class_weight=balanced

Trained on CIC-DDoS2019 (18 attack types)

Accuracy: ~100% · FPR: 0.01%

Logistic Regression

max_iter=1000 · class_weight=balanced

Trained on CIC-DDoS2019 (18 attack types)

Accuracy: 99.97% · FPR: 0.14%

Both models cover 18 attack types: SYN flood, UDP amplification, DNS reflection, LDAP, MSSQL, SNMP, TFTP, NetBIOS, NTP, SSDP, Portmap, WebDDoS, and more. The best-performing model result is shown in the dashboard.

GitHub	github.com/Huzaifal997/DDoS-Analyzer-Ai-Based-DDoS-Attack-Classification-
Dataset	CIC-DDoS2019 — Canadian Institute for Cybersecurity · unb.ca/cic/datasets/ddos-2019.html
Tool	CICFlowMeter — unb.ca/cic/research/applications.html

DDoS Analyzer · AI-Based DDoS Attack Classification · FYP Project